

Chapter 1: Analyzing Business Goals and Constraints

Using a Top-Down Network Design Methodology

According to Albert Einstein:

“The world we’ve made as a result of the level of thinking we have done thus far creates problems that we cannot solve at the same level at which we created them.”

To paraphrase Einstein, networking professionals have the ability to create networks that are so complex that when problems arise they can’t be solved using the same sort of thinking that was used to create the networks. Add to this the fact that each upgrade, patch, and modification to a network can also be created using complex and sometimes convoluted thinking, and you soon realize that the result is a network that is hard to understand and troubleshoot. A network created with this complexity often doesn’t perform as well as expected, doesn’t scale as the need for growth arises (as it almost always does), and doesn’t match a customer’s requirements. A solution to this problem is to use a streamlined, systematic methodology in which the network or upgrade is designed in a top-down fashion.

Many network design tools and methodologies in use today resemble the “connect-the dots” game. These tools let you place internetworking devices on a palette and connect them with LAN or WAN media. The problem with this methodology is that it skips the steps of analyzing a customer’s requirements and selecting devices and media based on those requirements. Good network design must recognize that a customer’s requirements embody many business and technical goals, including requirements for availability, scalability, affordability, security, and manageability. Many customers also want to specify a required level of network performance, often called a service level. To meet these needs, difficult network design choices and tradeoffs must be made when designing the logical network before any physical devices or media are selected.

When a customer expects a quick response to a network design request, a bottom-up (connect-the-dots) network design methodology can be used, if the customer’s applications and goals are well known. However, network designers often think they understand a customer’s applications and requirements only to discover, after a network is installed, that they did not capture the customer’s most important needs. Unexpected scalability and performance problems appear as the number of network users increases. These problems can be avoided if the network designer uses top-down methods that perform requirements analysis before technology selection.

Top-down network design is a methodology for designing networks that begins at the upper layers of the OSI reference model before moving to the lower layers. The top-down methodology focuses on applications, sessions, and data transport before the selection of routers, switches, and media that operate at the lower layers.

The top-down network design process includes exploring organizational and group structures to find the people for whom the network will provide services and from whom the designer should get valuable information to make the design succeed. Top-down network design is also iterative. To avoid getting bogged down in details too quickly, it is important to first get an overall view of a customer’s requirements. Later, more detail can be gathered on protocol behavior, scalability

requirements, technology preferences, and so on. Top-down network design recognizes that the logical model and the physical design can change as more information is gathered.

Using a Structured Network Design Process

Top-down network design is a discipline that grew out of the success of structured software programming and structured systems analysis. The main goal of structured systems analysis is to more accurately represent users' needs, which unfortunately often are ignored or misrepresented. Another goal is to make the project manageable by dividing it into modules that can be more easily maintained and changed.

With large network design projects, modularity is essential. The design should be split functionally to make the project more manageable. For example, the functions carried out in campus LANs can be analyzed separately from the functions carried out in remote-access networks, virtual private networks (VPN), and WANs. Cisco recommends a modular approach with its three-layer hierarchical model. This model divides networks into **core, distribution, and access layers**. With a structured approach to network design, each module is designed separately, yet in relation to other modules.

Network design is divided into four major phases that are carried out in a cyclical fashion:

- **Analyze requirements:** In this phase, the network analyst interviews users and technical personnel to gain an understanding of the business and technical goals for a new or enhanced system. The task of characterizing the existing network, including the logical and physical topology and network performance, follows. The last step in this phase is to analyze current and future network traffic, including traffic flow and load, protocol behavior, and quality of service (QoS) requirements.

- **Develop the logical design:** This phase deals with a logical topology for the new or enhanced network, network layer addressing, naming, and switching and routing protocols. Logical design also includes security planning, network management design, and the initial investigation into which service providers can meet WAN and remote access requirements.

- **Develop the physical design:** During the physical design phase, specific technologies and products that realize the logical design are selected. Also, the investigation into service providers, which began during the logical design phase, must be completed during this phase.

- **Test, optimize, and document the design:** The final steps in top-down network design are to write and implement a test plan, build a prototype or pilot, optimize the network design, and document your work with a network design proposal.

These major phases of network design repeat themselves as user feedback and network monitoring suggest enhancements or the need for new applications.

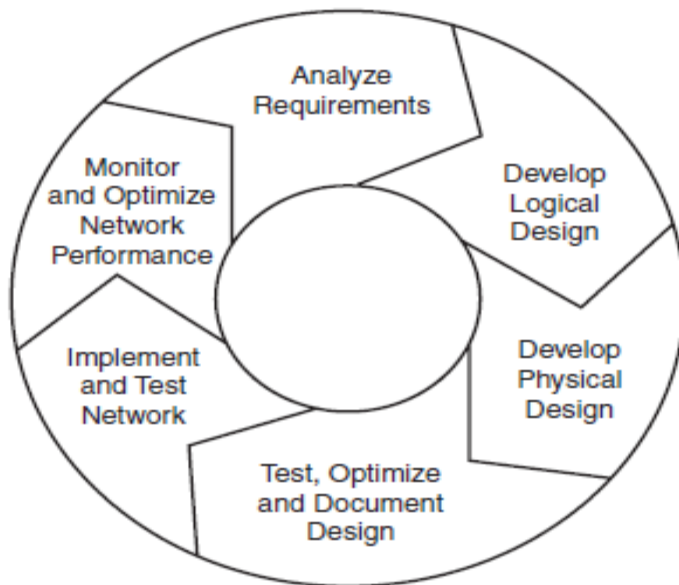


Figure 1-1 *Network Design and Implementation Cycle*

Plan Design Implement Operate Optimize (PDIOO) Network Life Cycle

The PDIOO life cycle includes the following steps:

- **Plan:** Network requirements are identified in this phase. This phase also includes an analysis of areas where the network will be installed and an identification of users who will require network services.
- **Design:** In this phase, the network designers accomplish the bulk of the logical and physical design, according to requirements gathered during the plan phase.
- **Implement:** The network is built according to the design specifications. Implementation also serves to verify the design.
- **Operate:** Operation is the final test of the effectiveness of the design. The network is monitored during this phase for performance problems and any faults to provide input into the optimize phase of the network life cycle.
- **Optimize:** The optimize phase is based on proactive network management that identifies and resolves problems before network disruptions arise. The optimize phase may lead to a network redesign if too many problems arise because of design errors or as network performance degrades over time as actual use and capabilities diverge.
- **Retire:** When the network, or a part of the network, is out-of-date, it might be taken out of production. Although Retire is not incorporated into the name of the life cycle (PDIOO), it is nonetheless an important phase. The retire phase wraps around to the plan phase. The PDIOO life cycle repeats as network requirements evolve.

Figure 1-2 shows a graphical representation of the Cisco PDIOO network life cycle.

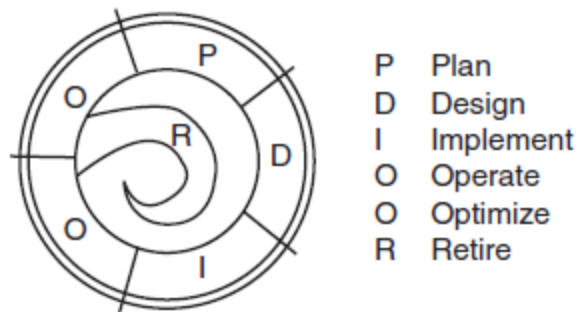


Figure 1-2 *PDIOO Network Life Cycle*

Analyzing Business Goals

Understanding your customer's business goals and constraints is a critical aspect of network design. Armed with a thorough analysis of your customer's business objectives, you can propose a network design that will meet with your customer's approval.

Working with Your Client

Before meeting with your customer try to have a clear understanding of the company and their business before you meet the customer. This will give you a clear understanding of the client's market and the role of the network in their business process. In your first meeting with your customers, ask them the following questions

- ✓ Explain the organizational structure of the company?
- ✓ What is the overall goal of the network design project?
- ✓ What is the business purpose of the network?
- ✓ Why does the customer want a new network design?
- ✓ For what will the new network be used?
- ✓ How will the new network help the customer be more successful in the customer's business?
- ✓ What goals must be met for the customer to be satisfied?
- ✓ Does the customer's definition of success will change as yearly fiscal goals change?
- ✓ What will happen if the network design project fails or if the network, when installed, does not perform to specification?
- ✓ How visible is the project to upper-level management?
- ✓ Will the success (or possible failure) of the project be visible to executives?
- ✓ To what extent could unforeseen behavior of the new network disrupt business operations?

In general, gather enough information to feel comfortable that you understand the extent and visibility of the network design project. You should try to get an overall view of whether the new network is critical to the business's mission.

Changes in Enterprise Networks

Enterprise networks at many corporations have been undergoing major changes. The main objective creating a network at the first place is to allow vast amount of information the company

has to be shared with **employees**, **customers**, and **business partners**, whenever and wherever this people are. A network that is used by only internal users is no longer the norm at many companies. Companies are seeking ways to build networks that more closely resemble modern organizations. Many modern organizations are based on an open, collaborative environment that provides access to information and services for many different constituents, including customers, prospective customers, vendors, suppliers, and employees. The network design must make sure the vast amount resources in the network are accessible from anywhere and anytime by authorized users and as the same time protected from unauthorized user that might harm the network.

Networks Must Make Business Sense

Although in the past many companies made “technology for technology’s sake” choices, this is no longer the case. Business leaders are more involved in Information Technology (IT) decisions than they once were, and IT managers rely on business managers to help them prioritize and fund IT projects. Network upgrades are made not because some new technology sounds interesting to the engineers, but because it will help an enterprise increase profits, productivity, market share, and cash flow. Network designers must choose solutions that address the business dilemmas faced by business managers.

The operational cost of the network has been reduced at some companies. Companies want the network services that will help them being productive and profitable as the same time want to reduce the operational cost of the network. You have to design a network that provides an efficient networking service with low operational cost. Companies are researching ways to make their data centers more efficient in their usage of power, cabling, racks, storage, and WAN circuits. Companies seek to reduce data center costs and to make data centers more “green” (whereby energy usage is reduced). Data center managers have discovered that many of their servers’ CPUs are underutilized. A major trend in enterprise network design is server virtualization, where one hardware platform supports multiple virtual servers. In the current business environment, security and disaster recovery should be considered with every network design choice, and the network designer must propose solutions that provide resiliency and stability.

Typical Network Design Business Goals

- ✓ List of some typical network design business goals:
- ✓ Increase revenue and profit
- ✓ Increase market share
- ✓ Expand into new markets
- ✓ Increase competitive advantages over companies in the same market
- ✓ Reduce costs
- ✓ Increase employee productivity
- ✓ Shorten product-development cycles
- ✓ Use just-in-time manufacturing
- ✓ Plan around component shortages
- ✓ Offer new customer services
- ✓ Offer better customer support

- ✓ Open the network to key constituents (prospects, investors, customers, business partners, suppliers, and employees)
- ✓ Avoid business disruption caused by network security problems
- ✓ Avoid business disruption caused by natural and unnatural disasters
- ✓ Modernize outdated technologies
- ✓ Reduce telecommunications and network costs, including overhead associated with separate networks for voice, data, and video
- ✓ Make data centers more efficient in their usage of power, cabling, racks, storage, and WAN circuits
- ✓ Comply with IT architecture design and governance goals

Identifying the Scope of a Network Design Project

One of the first steps in starting a network design project is to determine its scope. Some of the most common network design projects these days are small in scope for example, projects to allow a few people in a sales office to access the enterprise network via a VPN. On the other hand, some design projects are large in scope.

Ask your customer the following questions to understand the scope of the network design

- ✓ Ask if the design is for a single network segment
- ✓ Ask if the design is for the sets of LANs, a sets of WANs or remote access networks or the entire enterprise network
- ✓ Ask if the design is for the new network or a modification to existing one
- ✓ Explain to the customer about any concern you have about the scope of the project including technical and business concerns

Once you have the answer for the listed questions, double-check that you have gathered all the requirements and that you have accurate information about sites, links, and devices. If the project addresses network security, make sure you know about all external links, including any legacy dial-in access. When analyzing the scope of a network design, you can refer to the seven layers of the OSI reference model to specify the types of functionality the new network design must address.

In addition to using the OSI reference model, you can also use the following terms to define the scope of a network and the scope of a network design project:

- **Segment:** A single network bounded by a switch or router and based on a particular Layer 1 and Layer 2 protocol such as Fast Ethernet.
- **LAN:** A set of switched segments based on a particular Layer 2 protocol such as Fast Ethernet and an inter switch trunking protocol such as the IEEE 802.1Q standard.
- **Building network:** Multiple LANs within a building usually connected to a building backbone network.
- **Campus network:** Multiple buildings within a local geographical area (within a few miles), usually connected to a campus-backbone network.
- **Remote access:** Networking solutions that support individual remote users or small remote branch offices accessing the network.
- **WAN:** A geographically dispersed network including point-to-point, Frame Relay, ATM, and other long-distance connections.
- **Wireless network:** A LAN or WAN that uses the air (rather than a cable) for its medium.

■ **Enterprise network:** A large and diverse network, consisting of campuses, remote access services, and one or more WANs or long-range LANs. An enterprise network is also called an *internetwork*.

Identifying a Customer's Network Applications

At this point in the design process, you have identified your customer's business goals and the scope of the project. It is now time to focus on the real reason networks exist: applications. The identification of your customer's applications should include both current applications and new applications. Ask your customer to help you fill out a chart, such as the one in Table 1-1.

Table 1-1 *Network Applications*

Name of Application	Type of Application	New Application? (Yes or No)	Criticality	Comments

For Name of Application, simply use a name that your customer gives you. This could be an industry-standard name, such as Lotus Notes, or it could be an application name that means something only to the customer (especially for a home-grown application). For new applications, the name might be a code name for a software-development project. For Type of Application, you can use any appropriate text that describes the type of application, or you can classify the application as one of the following standard network applications:

- Email
- File transfer, sharing, and access
- Database access and updating
- Web browsing
- Network game
- Remote terminal
- Calendar
- Medical imaging
- Videoconferencing
- Video on demand (VoD)
- Scheduled multicast video
- Surveillance and security camera video
- Internet or intranet voice (IP telephony)
- Internet or intranet fax
- Sales order entry
- Management reporting
- Sales tracking
- Computer-aided design
- Document imaging
- Inventory control and shipping
- Telemetry
- Interactive Voice Response (IVR)

- Unified messaging
- Desktop publishing
- Web publishing
- Electronic whiteboard
- Terminal emulation
- Online directory (phone book)
- Distance learning
- Point of sales (retail store)
- Electronic commerce
- Financial modeling
- Human resources management
- Computer-aided manufacturing
- Process control and factory floor

In the Criticality column of the Network Applications chart, you can give each application a ranking from 1 to 3 with the following meanings:

1. Extremely critical
2. Somewhat critical
3. Not critical

Analyzing Business Constraints

In addition to analyzing business goals and determining your customer's need to support new and existing applications, it is important to analyze any business constraints that will affect your network design.

Politics and Policies

It has been said that there are two things not to talk about with friends: politics and religion. It would be nice if you could escape discussing office politics and technological religion (technology preferences) with a network design customer, but avoiding these topics puts your project at risk. In case office politics, your best bet is to listen rather than talk.

Learn about the inner working of the office politics by observing and listening to

- ✓ Learn about anyone who has a hidden agendas, turf wars, biases, group relations, or history behind the project that could cause it to fail
- ✓ Pay attention to personnel issues that could affect the project
- ✓ Determine which manager or managers started the project and how much do they have at stake?
- ✓ Are there any managers, network engineers, or users who want the project to fail for any reason? Find out who your advocates and opponents are.
- ✓ In some cases, no matter how technically sound your network design is, there will be people who have a negative reaction to it.
- ✓ Be sure to find out if your project will cause any jobs to be eliminated. Some network design projects involve automating tasks that were once done by highly paid workers. These workers obviously will have reasons to want the project to fail.
- ✓ Find out if there is a strategic business or IT plan.
- ✓ Does your network design need to fit into an overall architecture that is based on strategic planning?

- ✓ Are there external regulatory or governmental pressures on the planning process or on the architecture?

These sorts of pressures can often lead to messy political battles that can affect your network design. Be prepared for the possibility of formidable office politics if your network design project involves the merging of voice and data networks.

Budgetary and Staffing Constraints

Your network design must fit the customer's budget. The budget should include

- ✓ Allocations for equipment purchases,
- ✓ Software licenses, maintenance and support agreements, testing, training, and staffing.
- ✓ The budget might also include consulting fees (including your fees) and outsourcing expenses.

In general, it is a good idea to analyze the abilities of the networking staff. How much in house expertise is there? Should you recommend any training or outsourcing for network operations and management? The technologies and protocols that you recommend will depend on the abilities of internal staff. It is not a good idea to recommend a complex routing protocol, such as Open Shortest Path First (OSPF), for example, if the engineering staff is just starting to learn internetworking concepts (unless you also recommend a comprehensive training plan).

Project Scheduling

An additional business-oriented topic that you should review with your customer is the timeframe for the network design project. When the final due date and what is are the intermediate and major milestones? In most cases, management of the project schedule is the customer's obligation, not yours, but you should ask the customer to give you a copy of the schedule and to keep you informed about any slips in the schedule.

Business Goals Checklist

You can use the following checklist to determine if you have addressed your client's business-oriented objectives and concerns. If you can't gather every piece of data mentioned in the checklist, make sure you document what is missing in case it becomes critical, but don't stall the project to gather every last detail. Use the following business goal checklist

- ✓ I have researched the customer's industry and competition.
- ✓ I understand the customer's corporate structure.
- ✓ I have compiled a list of the customer's business goals, starting with one overall business goal that explains the primary purpose of the network design project.
- ✓ The customer has identified any mission-critical operations.
- ✓ I understand the customer's criteria for success and the ramifications of failure.
- ✓ I understand the scope of the network design project.
- ✓ I have identified the customer's network applications (using the Network Applications chart).
- ✓ The customer has explained policies regarding approved vendors, protocols, or platforms.
- ✓ The customer has explained any policies regarding open versus proprietary solutions.
- ✓ The customer has explained any policies regarding distributed authority for network design and implementation.

- ✓ I know the budget for this project.
- ✓ I know the schedule for this project, including the final due date and major milestones, and I believe it is practical.
- ✓ I have a good understanding of the technical expertise of my clients and any relevant internal or external staff.
- ✓ I have discussed a staff-education plan with the customer.
- ✓ I am aware of any office politics that might affect the network design.